CHALMERS

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Regulations for degree projects and Master's theses at first and second cycle levels at Chalmers

Policy document at Chalmers tekniska högskola AB

Contents

1. Background

These regulations constitute a revision of the *Regulations for Degree projects at Chalmers* (reg. no. 2024-1325), which are in turn an amalgamation of previous regulations for degree projects that replace:

- Instructions for theses on the MSc in Engineering, Master of Architecture and Master's programmes (reg. no. C 2016-0973)
- Regulations for degree projects for the Degree of BSc in Engineering at Chalmers University of Technology (reg. no. C 2019-1523)
- Regulations for independent projects (degree projects) on the Marine Engineering, Nautical Science, International Logistics and Shipping and Logistics programmes at Chalmers University of Technology (reg. no. C 2020-1531)

In addition, the following guiding principles for assessing the quality of degree projects and Master's theses are attached:

- Appendix 1 Guiding principles for assessing the quality of Master's theses on Chalmers' MSc in Engineering, Master of Architecture and Master's programmes. Included as an appendix in Regulations for theses on the MSc in Engineering, Master of Architecture and Master's programmes (reg. no. C 2016-0973)
- Appendix 2 Guiding principles for assessing the quality of degree projects on Chalmers' BSc in Engineering programmes. (reg. no. C 2011-893)
- Appendix 3 Guiding principles for assessing quality and grading for independent projects (degree projects) on the Marine Engineering, Nautical Science, International Logistics and Shipping and Logistics programmes at Chalmers University of Technology. (reg. no. C 2020-1532)
- Appendix 4 Guiding principles for assessing the quality of degree projects on Chalmers' Business Development and Entrepreneurship programme
- Appendix 5 Simple risk assessment template

Changes compared to the previous version are as follows:

- Updating the name of the document to: Regulations for degree projects and Master's theses at first and second cycle levels at Chalmers
- Adaptations to the course syllabuses of the degree projects and Master's theses. For specific course objectives, refer to the course syllabuses. Adjustments to the requirements for passing.
- Supplementation with special regulations for Master's theses in Architecture.
- The regulation has been supplemented by the degree project for Business Development and Entrepreneurship (TAFFS).
- The regulation has been supplemented with the purpose and learning objectives of the Master's programme in Learning and Leadership (MPLOL).
- The instructions for the planning report (7.3) have been supplemented with a requirement for a risk assessment, which must be attached to the report.

2. Degree project or Master's thesis

Chalmers' term for an independent project conducted within the framework of Master of Engineering, Architecture, or Master's programmes is *Master's thesis*.

Chalmers' term for an independent project conducted within the framework of BSc in Engineering programmes is *Degree project*.

Chalmers' term for an independent project conducted within the framework of Marine Engineering and Nautical Science programmes is *Degree project*. On the International Logistics and Business Development and Entrepreneurship programmes, the term is *Degree project*.

In order to simplify the writing in this document, both *Master's thesis* and *degree project* are referred to as degree project where reference is made to all types of independent project. If reference is made to just one of the types, it is specified.

3. Scope of the degree project

As an independent higher education provider, a foundation university, Chalmers must apply the Higher Education Ordinance (1993:100)'s Annex 2, System of Qualifications. According to this, the following applies to degree projects:

- For the Master of Architecture/MSc in Engineering degrees, the student must have completed an independent project (Master's thesis) of at least 30 higher education credits.
- For a Master's degree, the student must have completed an independent project (Master's thesis) of at least 30 higher education credits within the main field of study, as part of the course requirements. The independent project may be less than 30 higher education credits but must be at least 15 credits if the student has already completed an independent project of at least 15 higher education credits at third cycle level within the main field of study or equivalent abroad.
- For the BSc in Engineering, the student must have completed an independent project (degree project) of at least 15 higher education credits as part of the course requirements.
- For the BSc in Marine Engineering, the student must have completed an independent project (degree project) of at least 15 higher education credits as part of the course requirements.
- For the BSc in Nautical Science, the student must have completed an independent project (degree project) of at least 15 higher education credits as part of the course requirements.
- For the Bachelor's degree, the student must have completed an independent project (degree project) of at least 15 higher education credits within the main field of study as part of the course requirements.

For the MSc in Engineering, Master of Architecture, and Master's programmes at Chalmers, the Master's thesis is a course of either 30 or 60 higher education credits.

4. Purpose and learning objectives

The aim of a degree project is for students to develop in-depth knowledge, understanding, skills, and attitudes within the context of the education. A degree project should be carried out at the end of the education and involve a deepening and synthesis of previously acquired knowledge.

Master's thesis on the MSc in Engineering, Master of Architecture and Master's programmes

In the Master's thesis on an MSc in Engineering, Master of Architecture or Master of Science programme, the emphasis is on technical/scientific/artistic content. The overall objective of the Master's thesis is for the student to demonstrate the knowledge and skills required to work independently as an engineer/architect/Master of Science.

The learning objectives for Master's theses are based on the objectives for the MSc in Engineering, Master of Architecture and Master's degrees in the national system of qualifications and the local system of qualifications for first and second cycle levels applicable at Chalmers during the implementation of the thesis work. The specific learning objectives for a Master's thesis are set out in the course syllabus for the thesis in question.

The Master's thesis for the Master's programme in Learning and Leadership must also meet the requirements for a Master of Science in Secondary Education. Consequently, the thesis is expected to have both relevance to the engineering profession and a clear learning dimension. The specific learning objectives for a Master's thesis in Learning and Leadership are set out in the course syllabus for the thesis in question.

Degree projects on BSc in Engineering programmes

A degree project on a BSc in Engineering programme should reflect the nature of the education and demonstrate the student's ability to solve technical engineering problems.

The overall objective of the degree project is for the student to demonstrate the knowledge and skills required to work independently as a graduate with a BSc in Engineering.

The learning objectives for degree projects are based on the objectives for the BSc in Engineering degree in the national system of qualifications and the local system of qualifications for first and second cycle levels applicable in each case at Chalmers. The specific learning objectives for a degree project are set out in the course syllabus for the project in question.

Degree project for Business Development and Entrepreneurship

A degree project should reflect the nature of the education and demonstrate the student's ability to solve technical problems related to their profession.

The overall objective of the degree project is for the student to demonstrate that they have the knowledge and ability required to work independently in their professional area.

The learning objectives for degree projects are based on the objectives for the Bachelor's degree in the national system of qualifications and the local system of qualifications for first and second cycle levels applicable at Chalmers during the implementation of the thesis work. The specific learning objectives for a degree project are set out in the course syllabus for the project in question.

Degree projects in the International Logistics, Marine Engineering and Nautical Science programmes

The student should demonstrate the ability to critically and independently utilise, systematise and reflect on experiences and relevant research findings, thereby contributing to the development of professional practice and knowledge development within the profession, and reflect the nature of the education and demonstrate the ability to solve problems related to the profession.

The overall objective of the degree project is for the student to demonstrate that they have the knowledge and ability required to work independently in their professional area.

The learning objectives for the degree project are based on the objectives for the BSc in Nautical Science, BSc in Marine Engineering and Bachelor's degrees in the national system of qualifications and the local system of qualifications for first and second cycle levels applicable at Chalmers during the implementation of the thesis work. The specific learning objectives for a degree project are set out in the course syllabus for the project in question.

5. Examiner and supervisor

An examiner must be appointed for each degree project. The examiner is scientifically and qualitatively responsible for the degree project and for ensuring that the learning objectives are met. The examiner decides when the project can be approved and decides on the grade for the degree project.

The *Appointment regulation for teaching and research faculty* at Chalmers during the implementation of the thesis work sets out who may be appointed as an examiner. Each department is responsible for ensuring that an examiner is appointed.

An examiner may appoint one or more supervisors. The supervisor provides scientific/technical/artistic support and assists the student(s) with the practical processes.

For the International Logistics, Nautical Science and Marine Engineering programmes, students choose supervisors within their chosen educational area, after which the department appoints an examiner for each group.

If the degree project is carried out in an external organisation, students typically also have a supervisor in the organisation in which the degree project is carried out. This supervisor is referred to as an external supervisor.

6. Conditions for starting a degree project

To start a degree project, the following higher education credit requirements must be met in each programme:

- MSc in Engineering/Master of Architecture - at least 225 higher education credits.
- Students admitted only to a Master's programme - at least 45 higher education credits.
- BSc in Engineering programme - at least 120 higher education credits.
- Degree project for Business Development and Entrepreneurship - at least 120 higher education credits.
- International Logistics, Nautical Science and Marine Engineering programmes - at least 120 higher education credits. Higher education credits from ship-based education are not included. In the case of alternated theory and practice in the Nautical Science and Marine Engineering programmes, at least 180 days and 150 days of ship-based education, respectively, are required (half of the total practical time).

In addition to meeting the general requirements, the necessary prerequisite courses for the specific degree project must have been completed. The examiner formulates and verifies such prerequisites.

7. Implementation

A degree project is carried out by one student or two students together.

7.1 Initiation

A degree project is usually initiated in one of the following ways:

- The student contacts a company or a department with a proposal for a degree project. At the same time, the student contacts an examiner in the appropriate subject area at Chalmers or the person responsible for degree projects at the department.
- A proposal from a company or department on the thesis portal (annonsportal.chalmers.se).

7.2 Preparatory administration

The student must write a brief description of the intended degree project. The description must provide sufficient information for the examiner to decide whether the task is suitable for a degree project. The description must include the background, purpose, objectives and possibly methods. The description may, if relevant, also be written in consultation with the external company/organisation where the degree project is to be carried out. It is possible to attach this description in the online form below. Otherwise it must be emailed directly to the examiner.

For Master's theses in **Architecture**, the student must have an approved project plan in accordance with the instructions in the course syllabus for Master's theses before the thesis may be started. The project plan must be submitted in the online 'Thesis application form' according to the timetable indicated.

In some cases, the 'Thesis application exemption form' must be used. This exemption form can be accessed via chalmers.se. This should only be used:

- If the main field of study of the course does not correspond to the main field of study of the programme (applies only to degree projects of 30 or 60 higher education credits)
- If the student wants to register for an older programme that is no longer offered.

The form for the degree project (online/exemption form) must first be digitally signed by the examiner, who thereby assesses and approves that the implementation of the proposed degree project leads to the student(s) developing the knowledge, abilities and attitudes that are part of the degree project's learning objectives.

After the examiner has signed one of the two forms, the Head of Programme/Director of Master's Programme must make an assessment.

For students on a Master's programme, the Director of Master's Programme (MPA) must assess and approve that the proposal falls within the main field of study of the Master's programme. The MPA may decide that a specific thesis belongs to the main field of study of the Master's programme, even if the department at which the thesis is to be carried out does not belong to the main field of study according to decision C2008/280.

If the student(s) is (are) admitted to the programmes in Engineering/Architecture, the MPA must also assess and approve that the thesis is relevant in terms of the technical area/artistic area. The MPA then signs the form (online/exemption form).

For students on the BSc in Engineering, Marine Engineering, Nautical Science and Bachelor's programmes, the Head of Programme (PA) checks that the project falls within the technical area/main field of study of the programme. The examiner assesses and approves that the implementation of the proposed degree project leads to the student(s) developing the knowledge, abilities and attitudes that are part of the degree project's learning objectives and that the project is assessed as corresponding to 15 higher education credits. The PA then signs the form (online/exemption form).

Information from the online form will be transferred to Ladok after the form has been signed, first by the examiner and then by the PA/MPA. The student can then register.

(If individual dates are selected or if something else is wrong, a copy of the online form will automatically be sent to thesis@chalmers.se , and Thesis Admin. will handle the case and process it according to standard procedures).

The exemption form must, after it has been signed, first by the examiner and then by the PA/MPA, be sent to: thesis@chalmers.se, which will transfer the information to Ladok. The student can then register.

Work card

The student(s) fill in a digital work card (pdf) that is downloaded from Chalmers.se. After each completed mandatory part, the student(s) ensures(ensure) that this is signed on the work card.

Digital work cards are not used at some departments. Information about what applies to degree projects on these programmes can be found in Canvas.

7.3 Planning report

The student(s) must write a planning report that specifies the problem description/task. The planning report must include the background, preliminary purpose, objectives, limitations and method, and a timetable for the degree project's implementation. In the planning report, the student(s) must highlight the societal, ethical and ecological aspects that need to be considered according to the learning objectives. If such aspects are not considered, reasons for not doing so must be given. All degree projects must also be risk assessed using the 'Simple risk assessment template'. A students carry out the risk assessment with the examiner and any external supervisor. The risk assessment must be attached to the planning report and approved by the examiner. More information about the examiner's health and safety responsibilities can be found on the intranet and the page *'Teacher's health and safety tasks'*.

For a degree project comprising 60 higher education credits, an interim objective for 30 higher education credits must be specified.

The planning report must be approved on the work card by the examiner.

For Master's theses in **Architecture**, the planning report is replaced by the more detailed project plan, which must be approved before the Master's thesis may be started.

7.4 Supervision

During the project, the student(s) are entitled to regular supervision and other resources needed for the project's implementation.

Supervision is provided by Chalmers' supervisors and possibly external supervisors. An external supervisor continuously supports students during the project's implementation and assists them with practical processes at the external organisation.

7.5 Interim presentations

For Master's theses in **Architecture**, a presentation is made at a midterm seminar.

Students who carry out degree projects of 60 higher education credits must report on the status of the project to the examiner after twenty working weeks. Half of the work required to achieve the learning objectives for a degree project of 60 higher education credits should then have been completed. An approved interim presentation means that 30 higher education credits are reported in Ladok.

For degree projects in the **International Logistics, Marine Engineering and Nautical science programmes**, a midterm seminar is held in the middle of the course. The purpose of the midterm seminar is to clarify progress and receive feedback from the examiner.

7.6 Public access and confidentiality

In accordance with Chalmers' Appointment regulation for first cycle and second cycle education (undergraduate and Master's education), a degree project must be presented openly both in writing and orally. The entire essay must be public. The rule on public access also applies to essays that are not published in full (see also section 8.3 on e-publication).

In other respects, Chalmers' policies on public access and confidentiality apply to the implementation of a degree project and publication of the essay.

7.7 Copyright

Legislation on copyright has its origins in the Swedish Act on Copyright in Literary and Artistic Works (1960:729), normally referred to as the Copyright Act. Copyright consists of an economic right and a moral right that both accrue to the author. The author is the natural person who created the work, i.e. the student(s).

The author may choose to enter into an agreement to transfer the economic rights to others, either as a whole or in part, or give others the right to use them. Transfer means that the author waives their rights and hands over ownership of the economic rights on agreed terms.

The moral right comprises the author's right to be named as the author of a work and the right not to have to accept changes being made to their work or the work being published in contexts that may be prejudicial to their artistic or literary reputation or unique character.

8 Examination

8.1 Grades

For Master's theses on the MSc in Engineering, Master of Architecture and Master's programmes, and degree projects in Business Development and Entrepreneurship and on the BSc in Engineering programme, grades are awarded on the UG scale with grades U (Fail) and G (Pass).

For degree projects on the International Logistics, Nautical Science and Marine Engineering programmes, grades are awarded on the TH scale, with grades U (Fail), 3 (Pass), 4 (Pass with Distinction) and 5 (Pass with Great Distinction).

To pass, the following is required:

- Approved planning report

- Approved presentation and defence on the degree project presentation
- Approved opposition of another degree project
- Attendance at two other presentations
- Approved report under prevailing standards for scientific and technical reporting

For degree projects on the **International Logistics, Nautical Science and Marine Engineering programmes**, the following are also required:

- Approved midterm seminar
- Approved sections on information and communication skills

For degree projects on **BSc in Engineering programmes** and **Business Development and Entrepreneurship**, the following is also required:

- Approved sections on information and communication skills

For Master's theses in **Architecture**, the following are required:

- Approved project plan
- Approved Master's thesis
- Presentation of the Master's thesis at the midterm seminar
- Approved presentation and defence of the Master's thesis at the final seminar
- Approved opposition of another Master's thesis at the midterm seminar and attendance at at least four additional presentations
- Presentation of the Master's thesis at the exhibition of Master's theses
- Submission of a report, digitally and in two printed copies

8.1.1 Criteria for the grade Pass for degree projects comprising 30 higher education credits

For degree projects comprising 30 higher education credits, a Pass grade requires that they at least meet the criteria for High Quality in respect of all learning objectives. To meet the criteria for High Quality in respect of learning objective 5 (see appendix with assessment criteria), the student must have passed all components listed above within a total time frame of 30 working weeks. The examiner may, for special reasons, extend this time limit by 10 working weeks at a time.

8.1.2 Criteria for the grade Pass for degree projects comprising 60 higher education credits

For degree projects comprising 60 higher education credits, a Pass grade requires that they meet the criteria for Very High Quality in respect of learning objectives 1 and 2, and at least High Quality for the other learning objectives (3-11). See appendix with assessment criteria. To meet the criteria for High Quality in respect of learning objective 5, the student must have passed all components listed above within a total time frame of 50 working weeks. The examiner may, for special reasons, extend this time limit by 10 working weeks at a time.

8.2 Written Presentation

The essay for a Master's thesis on the MSc in Engineering, Master of Architecture and Master's programmes must be written in English. Exceptions may only be made for Master's programmes taught in Swedish.

The essay for a degree project on the BSc in Engineering programme and the Business Development and Entrepreneurship programme must normally be written in Swedish.
In exceptional cases, it may be written in English.

For the International Logistics, Marine Engineering and Nautical Science programmes, the essay may be written in Swedish or English.

Essays must be formatted according to the template for formatting degree projects, which can be accessed at Chalmers.se. For Master's theses in **Architecture**, a different template is used according to specific instructions. When two students carry out a project together, the division of labour must be clearly outlined in a contribution report which is attached separately.

The essay must provide sufficient basis for the examiner to decide on a grade.

The examiner checks the essay against a plagiarism tool.

E-publication

Chalmers degree projects must be registered and published electronically in Chalmers' e-publication system. They are then searchable in the Student Theses service and freely available and searchable online. A student may decline electronic publishing, but the registration (i.e. a searchable record without full text) is mandatory. For electronic publication of the full text, all authors must have signed and approved the publication agreement on the work card.

Student theses and projects are registered and published electronically by the respective departments, and the examiner is responsible for ensuring that this is done.

8.3 Oral presentation

At the time of the oral presentation, the essay must have been completed but not yet published. This makes it possible for feedback received during the presentation to be incorporated in the essay.

The oral presentation, including opposition, must take place at Chalmers. Additional presentations may be held at the company, if required.

In exceptional cases, for example if the degree project was carried out abroad, the examiner may allow the presentation and opposition not to be held on site at Chalmers. In these cases, the presentation must be streamed digitally and be open to the public.

The presentation of the degree project should be advertised at the relevant department at least two weeks before the presentation date. The presentation should normally take place in the period from 15 August to 15 June and during normal working hours.

The oral presentation begins with the student(s) presenting their project. This is followed by opposition and discussion.

The presentation should take 45-60 minutes, with about one-third of the time devoted to opposition and discussion.

For Master's theses on the MSc in Engineering, Master of Architecture, and Master's programmes, the oral presentation must be in English. Exceptions may be granted for programmes taught in Swedish.

For degree projects in Marine Engineering, Nautical science, Bachelor's and BSc in Engineering programmes, the presentation should be in Swedish.

For Master's theses in **Architecture**, the thesis is assessed at the final seminar by a jury consisting of one external and one internal critic.

8.4 Opposition

A student must have participated as an opponent for another degree project. The department decides how the opposition is organised. A maximum of two students may act as opponents for the same degree project. Opponents have 10 minutes at their disposal, and all this time should be utilised. After the opponents have presented their comments, others present may ask questions.

Opponents must review the essay. Linguistic errors and minor remarks must be noted in writing and submitted after the opposition.

The opponents' efforts are assessed by the examiner for the project presented, and approved opposition is signed for on the work card.

The student(s) appoints(appoint) an opponent(opponents) for their own project.

For Master's theses in **Architecture**, other students' theses are opposed in connection with the midterm seminar.

On the **International Logistics, Nautical Science and Marine Engineering programmes**, opponents are appointed by the examiner.

8.5 Attendance at other presentations

A student must attend two other degree project presentations. The examiner for the degree project presented signs for approved attendance on the work card.

For Master's theses in **Architecture**, other requirements apply. See 8.1.

8.6 AI tools

The student(s) is(are) responsible for ensuring that the essay follows Chalmers' guiding principles and rules regarding academic honesty and the use of AI tools. See Chalmers' education pages. The examiner is responsible for having the essay/report checked in a plagiarism tool. An essay/report that contains plagiarism will not be awarded a pass. If an essay does not follow Chalmers' guiding principles and rules regarding academic honesty and the use of AI tools, it may lead to disciplinary measures according to Chalmers' Rules of Discipline.

9 Checklist for roles and responsibilities

The following section describes the responsibilities associated with different roles during the degree project. In addition, there may be local variations in work procedures and actual implementation at the department.

9.1 Examiner

It is the examiner's responsibility to:

- Assess and approve the scientific and qualitative level of the degree project and to ensure that the implementation of the degree project leads to the student(s) developing the knowledge, skills and attitudes included in the degree project's learning objectives.

- Inform themselves about *Guiding principles for Chalmers' collaboration with trade and industry*. (C 2007/884) when collaborating with companies.
- Possibly appoint supervisors and provide necessary guidance for supervision, both for the internal supervisor at Chalmers and external supervisors.
- Ensure that the degree project can be carried out using the given resources.
- Verify that the student(s) meets(meet) the prerequisites and credit requirements.
- Approve the thesis application form or the thesis application exemption form.
- Approve the planning report. For **Architecture**: Approve the project plan.
- Inform the student(s) of Chalmers' policies for public access and confidentiality.
- Ensure that the student(s) are offered regular supervision.
- Approve the interim presentation after 20 working weeks (equivalent to 30 higher education credits) for degree projects comprising 60 higher education credits, and ensure that the approved presentation is reported in Ladok.
- Check the essay with a plagiarism tool and take action if there is suspicion of plagiarism.
- Review the degree project and decide when the project can be presented.
- Ensure that the project is presented orally according to the applicable rules.
- Chair the presentation.
- Approve opposition to the degree project.
- Approve attendance at the presentation session.
- Sign all approved subtasks and approve the degree project when all subtasks have been approved.
- Ensure that the essay is registered and published electronically in Chalmers' e-publication system and as full text if the student has approved this.
- Ensure that the degree project is registered and archived.

9.2 Director of Master's Programme (MPA)/Head of Programme (PA)

The MPA is responsible for:

- Checking that the degree project falls within the main field of study of the Master's programme.
- Checking and approving that the Master's thesis belongs to the relevant technical area (equivalent) if the student intends to take the MSc in Engineering/Master of Architecture degree. The MPA may decide that a specific Master's thesis belongs to the main field of study of the Master's programme, even if the department at which the thesis is to be carried out does not belong to the main field of study according to decision C2008/280.
- Signing the thesis application form or the thesis application exemption form.

For BSc in Engineering, Bachelor's and Marine/Nautical programmes, the PA is responsible for:

- Checking and approving that the degree project belongs to the relevant main field of study/technical area.
- Signing the thesis application form or the thesis application exemption form.

9.3 Supervisor

The supervisor is responsible for:

- Providing ongoing scientific/technical/artistic support to the student(s) during the degree project and assisting them with practical processes.

For BSc in Engineering and Marine/Nautical programmes, the supervisor is also responsible for:

- Supervising the planning report for the degree project.
- Supporting the student(s) in completing the essay.
- Reviewing the degree project and assessing whether the essay is of sufficient quality for submission to the examiner.

9.4 Student

The student(s) is(are) responsible for:

- Finding a suitable issue for a degree project at a company or department.
- Describing the proposal for the degree project independently and in writing.
- Contacting the relevant department and examiner for the subject.
- Filling in the thesis application form or the thesis application exemption form.
- Ensuring that the work card is signed (does not apply to Marine/Nautical programmes).
- Planning, carrying out and presenting the project independently according to the requirements described above.
- Reading Chalmers' policies for public access and confidentiality before starting the project.
- Reading Chalmers' information on degree project essays and formatting the essay in accordance with Chalmers' rules for 'Formatting degree projects' and for publication in Chalmers' e-publication system.
- Contacting the examiner before any agreement on the degree project.
- Contacting and appointing opponent(s) for the presentation of their own work.
- Delivering the project for publication in Chalmers Open Digital Repository.
- When attending the presentation of another degree project, ensuring that the examiner in question knows that they were present.

9.5 Student and Education Office

The Student and Education Office is responsible for the degree project being established in Ladok.

APPENDIX 1. Guiding principles for assessing the quality of Master's theses on Chalmers' MSc in Engineering, Master of Architecture and Master's programmes

This appendix includes the guiding principles in the document 'Guiding principles for assessing the quality of Master's theses on Chalmers' MSc in Engineering, Master of Architecture and Master's programmes', reg. no. C2011/895.

The guiding principles are based on the learning objectives for the MSc in Engineering, Master of Architecture and Master of Science degrees, which are in Chalmers' local system of qualifications.

Ratings are given on a three-point scale: Poor Quality (BK), High Quality (HK), and Very High Quality (MHK).

Criteria for Very High Quality (MHK) are formulated only for specific learning objectives, those that are judged to be of a distinctive nature: significant specialisation in the main field of study, specialised method knowledge, problem formulation, ability to create and evaluate new solutions, written presentation and independence.

Learning objectives with guiding principles for quality criteria

The criteria for Poor Quality (BK), High Quality (HK), and Very High Quality (MHK) are presented below for each learning objective for Master's theses.

1. **Apply significantly specialised knowledge in the main field of study/specialisation of the programme in your project and relate it to current research and development work in a scientifically correct manner**

MHK	Significant specialisation in the main field of study is demonstrated. The thesis utilises knowledge from advanced-level studies in the main field of study. An extensive review of existing literature, along with reflection on the thesis' connection to the forefront of knowledge in the main field of study, is present. The thesis contributes to new knowledge in the main field of study in a clearly presented manner. The thesis demonstrates the ability to make an independent contribution to the field.
HK	Significant specialisation in the main field of study is demonstrated. The thesis utilises knowledge from advanced-level studies in the main field of study. A written review of existing literature, along with reflection on the thesis' connection to the forefront of knowledge in the main field of study, is present.
BK	The thesis' connection to the main field of study is weak and absent. Advanced-level knowledge is not utilised. A literature review and reflection on the thesis' connection to the relevant knowledge area are absent.

2. **Choose (and justify the choice) the method in the project, in the main field of study/specialisation of the programme**

MHK	Potentially relevant engineering or scientific theories and methods have been identified. A well-justified choice of theory and method has been made. The theories and methods chosen have been applied correctly and innovatively. The thesis demonstrates deep, broad understanding of methodology.
HK	Potentially relevant engineering or scientific theories and methods have been identified. A well-justified choice of theory and method has been made. The methods chosen have been correctly applied.
BK	The theories and methods chosen for the thesis lack relevance. The student has not shown mastery of the theories and methods chosen.

3. **Contribute to research and development work, and be able to relate your work to relevant scientific and technical/industrial/architectural contexts**

HK	The contribution to research and development work is clearly presented
BK	The nature of the thesis is such that it is difficult to link it to research and development work

4. **Identify, formulate and handle complex questions with a holistic view, critically, independently and creatively**

MHK	The thesis has a clear, distinct question or set of objectives. The question/objectives have been processed in an adequate, critical and reflective way. There is a clear link between the question/objectives, results, discussion and conclusions. The thesis' conclusions are well substantiated and correct.
HK	The thesis has a clear, distinct question. The question has been adequately processed. There is a clear connection between the research question, results and conclusions. The thesis' conclusions are well substantiated and correct.
BK	The thesis lacks or has an unclear question or set of objectives. Irrelevant methodology has been employed. The project does not provide an answer to the question or a result that relates to the objective. The conclusions are incorrect.

5. **Plan and, using adequate methods, carry out advanced tasks within given frameworks, and be able to evaluate this work**

HK	A realistic plan has been drawn up for the thesis. The deadlines that were communicated and established have been adhered to during the implementation of the thesis. Any adjustments that have been necessary for the implementation have been documented and communicated.
BK	The thesis has not adhered to the deadlines communicated and established, nor has documentation of relevant factors for deviations been presented.

6. **Create, analyse and critically evaluate different technical/architectural solutions**

MHK	New solutions are developed in the thesis that are analysed and evaluated in a critical way. Alternative solutions have been developed and processed in a relevant, exhaustive manner.
HK	Solutions are developed in the thesis that are analysed and evaluated in a critical way.
BK	The work has not presented the above in a clear way.

7. **Integrate knowledge critically and systematically**

MHK	The thesis integrates knowledge and methods from several subjects in an innovative way.
HK	Relevant knowledge and methods have been acquired and applied.
BK	Areas of relevance to the thesis are not addressed or used. Knowledge that has been chosen and acquired is not presented clearly and lacks justification.

8. **Present and discuss conclusions, and the knowledge and definitions on which they are based, clearly in oral and written English**

MHK	A very well-written essay. The overall coherence, structure, and layout are of very high quality.
HK	The thesis addresses the chosen area with relevant, accurate use of language. The overall coherence, structure and layout are of good quality.

BK	The thesis primarily lacks adequate use of language, making it difficult to understand or assess the thesis based on the report.

9. **Within the scope of the specific project, identify the questions that need to be answered to consider relevant societal, ethical and ecological aspects**

HK	Presents and justifies chosen methods and discusses results from a perspective with a focus on sustainable development.
BK	Does not consider this aspect. There is no justification provided in the planning report.

10. **Consider and discuss ethical aspects of research and development work, in terms of both *how* the work is carried out and *what* is investigated/developed.**

HK	Presents possible ethical consequences of the work performed.
BK	Does not consider this aspect. There is no justification provided in the planning report.

11. **Identify and discuss the need for further clarification of various aspects of the project before decisions are made or implementation, where relevant.**

HK	The student has reflected on and presented the other aspects that need to be clarified/investigated before decisions are made or implementation.
BK	The student has not critically considered which other aspects need to be clarified/investigated before decisions are made or implementation.

Overall objectives

On completion of the Master's thesis, the student should have demonstrated the knowledge and skills required to work independently as an engineer/architect/Master of science

MHK	A Master's thesis implemented independently and without extraordinary support or adjustments, or any other significant extra resources for its implementation.
HK	Carried out the thesis with reasonable support.
BK	There has been a great need for support. This support has been too extensive to make it likely that the student will be able to work independently after graduation.

APPENDIX 2. Guiding principles for assessing the quality of degree projects on Chalmers' BSc in Engineering programmes

The guiding principles are specifically intended to be a guide in cases in which the quality of the degree project is low, and there is reason to consider whether it can be passed. The guiding principles may also be a support when giving feedback to the student on their performance.

The guiding principles are based on the learning objectives for degree projects in BSc in Engineering programmes.

Criteria for poor quality/high quality are formulated for all learning objectives. Criteria for Very High Quality (MHK) are formulated only for specific learning objectives, those that are judged to be of a distinctive nature: specialisation in the technical area, problem formulation, ability to create and evaluate new solutions, integration of knowledge, written presentation and independence.

Poor quality in terms of a learning objective or several learning objectives in combination may result in a fail grade for the entire project. It is the examiner's responsibility to make a balanced assessment.

Criteria for Poor Quality (BK), High Quality (HK) and Very High Quality (MHK) for each of the learning objectives for the degree project are shown below.

1. **Ability to acquire and apply specialised knowledge in the technical area of the study programme, including specialised insight into current development work:**

MHK	Specialised knowledge in the technical area is demonstrated. The project contributes, in a clearly presented way, to new knowledge or new application in the technical area. The project demonstrates the ability to work independently in the area.
HK	Specialised knowledge in the technical area is demonstrated. The project shows the ability to work independently in the area.
BK	The project does not utilise specialised knowledge in the technical area. The project's connection to the technical area is weak or absent.

2. **The ability to independently and creatively identify, formulate, and manage questions with a holistic perspective, and analyse and evaluate various technical solutions:**

MHK	The thesis has a clear, distinct question or set of objectives. The question/objectives have been processed in an adequate, critical and reflective way. New solutions are developed and analysed and evaluated critically in the project. Alternative solutions have been developed and processed in a relevant, exhaustive manner. There is a clear connection between question/objectives, results, discussions and conclusions. The project's conclusions are well substantiated and correct.
HK	The project has a clear, distinct question. The question has been adequately processed. Solutions are developed, analysed and evaluated critically in the project.

	The project’s conclusions are well substantiated and correct. There is a clear connection between question, objectives, results and conclusions.
BK	The project lacks or has an unclear question or set of objectives. Irrelevant methodology has been employed. Alternative solutions have not been identified. The project does not provide an answer to the question or a result that relates to the objective. The conclusions are incorrect.

3. Ability to plan and carry out tasks with adequate methods within given frameworks:

HK	A realistic plan has been drawn up for the project. The deadlines that were communicated and established have been adhered to during the implementation of the thesis. Any adjustments that have been necessary for the implementation have been documented and communicated.
BK	The project has not adhered to the deadlines communicated and established. Furthermore, documentation of relevant factors for deviations has not been provided.

4. Ability to critically and systematically apply knowledge, and to model, simulate, predict and evaluate phenomena based on relevant information:

MHK	The project integrates knowledge and methods from several subjects.
HK	Relevant knowledge and methods have been acquired and applied.
BK	Areas of relevance to the project are not addressed or used. Knowledge that has been chosen and acquired is not presented clearly and lacks justification.

5. Ability to give an account of and discuss information, problems and solutions orally and in writing:

MHK	A well-written report. The overall coherence, structure and layout are of high quality.
HK	The project addresses the chosen area with relevant, accurate use of language. The overall coherence, structure and layout are of good quality.
BK	The project primarily lacks adequate use of language, making it difficult to understand or assess the project based on the report.

The guiding principles for the assessment of degree project reports can provide guidance for the assessment of the written report. See the document Assessment of the written presentation, report - HISS (version 2011-01-10).

http://www.ait.gu.se/facksprak/kandidatarbete/skrivanvisningar/

6. **Ability to identify, within the scope of the specific degree project, questions regarding the role of technology in society, such as environmental and ethical aspects:**

HK	Presents and justifies chosen methods and discusses results from a perspective with a focus on sustainable development. Presents possible ethical consequences of the project performed.
BK	Does not consider these aspects even though the examiner deems them to be of importance for the degree project in question.

This learning objective may be irrelevant for certain degree projects. This assessment is made by the examiner.

7. **After having completed the degree project, the student must have demonstrated the knowledge and skills required to work independently as a graduate engineer:**

MHK	A degree project implemented independently and without extraordinary support, or any other significant extra resources for its implementation.
HK	Implemented the project with reasonable support.
BK	There has been a great need for support. This support has been too extensive to make it likely that the student will be able to work independently after graduation.

APPENDIX 3. Guiding principles for assessing the quality of and grading degree projects on Chalmers' International Logistics, Nautical Science and Marine Engineering programmes.

Background

This document describes the criteria that apply for assessing the quality of and grading independent projects (degree projects) on the International Logistics, Nautical Science and Marine Engineering programmes at Chalmers University of Technology.

Assessment of degree projects

Grading is done according to the TH scale, i.e. with the grades Pass with Distinction (5), Pass with Credit (4), Pass (3) and Fail (U).

A student with a degree project that does not meet the criteria for a pass but that, according to the examiner's assessment, could do so with supplementary work, is given such an opportunity. Supplementary work involves the students having the possibility to address shortcomings in the project and subsequently to be awarded a grade 3. Supplementary work that is not completed within six calendar months from the date of submission of the project results in a fail grade.

A fail grade (U) means that the degree project is of such low quality that the students cannot be awarded a pass for their degree project and must start again with a new project.

The assessment consists of nine assessment criteria, each of them assigned to the three levels Very High Quality (MHK), High Quality (HK) and Poor Quality (BK). To achieve the Very High Quality (MHK) and High Quality (HK) levels, all the criteria for that level must be fulfilled.

The grade is given based on the number of MHK, HK and BK according to the table.

Learning Objectives for degree projects and link to assessment criteria

The overall objective of the degree project is for the student to demonstrate that they have the knowledge and ability required to work independently in their professional area.

Specific learning objectives for the degree project include the student's ability to:

1. acquire and apply specialised knowledge in the subject area of the educational programme, including specialised insight into current development work,
2. apply a comprehensive approach to identifying, formulating and managing research questions independently and creatively and analysing and evaluating them at a specialised level in the subject area,
3. plan and use appropriate methods to perform, analyse and evaluate tasks within specific frameworks and present theoretical and methodically sound arguments,
4. identify appropriate methods, apply them and critically evaluate them in relation to a chosen scientific question,
5. critically and systematically make use of knowledge and model, simulate, predict and evaluate phenomena based on relevant information,
6. within the framework of the specific degree project, identify research questions relating to the role of technology in society taking into account environmental and ethical aspects, and
7. report orally and in writing, for a given audience, discuss information, problems and solutions with high standards of structure, formalities and use of language, and defend an academic study.

The following table matches the seven learning objectives to the nine assessment criteria:

Learning objective	Checked against assessment criterion/criteria
1	3
2	1, 2
3	7, 8
4	7
5	8
6	7, 8
7	4, 5, 6, 7, 8, 9

Assessment criteria

The assessment is based on the following nine criteria.

The criteria for specialisation in the project have been assessed to be such an important part that they earn double MHK or BK points towards the grade. The process criterion has no BK as an inadequate process cannot be supplemented afterwards.

Process

<table>
<tr><td colspan="2">The contact between the supervisor and the student is extremely important in the course of the project. The supervisor provides ongoing support to students while they are working on their project and helps them with the practical processes. The scope and means of communication of the supervision arrangement must be agreed when the project begins.

A realistic project plan must be drawn up, which must include a breakdown of activities and a timetable. In the course of the project, a project diary and a time log must be kept.

The students are expected to implement their degree project independently, without too much extensive support by the supervisor, and to follow the normal degree project process.

NB! The assessment criteria are assessed by the supervisor!</td></tr>
<tr><td>MHK</td><td>a) Contact with the supervisor has been good and in line with what was agreed.

b) A realistic plan has been drawn up for the project. The deadlines communicated and established have been followed during the implementation of the project. Any adjustments that have been necessary for the implementation have been documented and communicated.</td></tr>
</table>

	c) A degree project conducted *independently* and without extraordinary support or adjustments, without requiring any other significant extra resources for its execution.
HK	a) Contact with the supervisor has been *good*. b) A realistic plan has been drawn up for the project. The deadlines communicated and established have *largely* been *followed* during the implementation of the project. c) Project implemented with *reasonable support*.
BK	a) Contact with the supervisor has been *inadequate*. b) The project has *not followed* the deadlines that were communicated and established, nor has it been possible for any documentation of relevant factors for these deviations to be presented. c) There has been *extensive need for support*. This support has been too extensive to make it likely that the student will be able to work independently after graduation.

Overall impression

For the project's overall impression to be good, the various parts of the report must cohere with one another, there must be a common thread that runs through the project and the subject description must be good. The report must follow current templates and be well balanced as well as being well written using good language and layout. The overall impression is something that can be assessed in a relatively short time when the text is first read through and on the basis of the impression it actually gives then. This impression, which also comprises understanding of the subject and correctness, is then confirmed or re-evaluated on a more thorough review of the text.	
MHK	a) The subject description is *excellent,* and the analysis of the results is comprehensible and coherent in terms of purpose and conclusion. Relevant areas are covered and described in a *clear* manner. In addition, their selection and scope are *clearly linked* to the purpose and general content of the study in a well-balanced report. b) The conclusions are clearly linked to and *highly consistent* with the analysis of results and the purpose/problem formulation. c) The research question/objectives have been formulated following adequate, *critical and reflective* consideration. There is a clear connection between the research question/objectives, results, discussion and conclusions. The project's conclusions are well substantiated and correct.

	d) Relevant knowledge and methodology have been *acquired, applied and justified.* e) A very *well-written* report. The overall coherence, structure and layout are of *very high* quality. f) The report contains a summary that reflects the content of the whole report *clearly and in a manner that arouses interest.*
HK	a) The subject description is *good*. Many important areas are mentioned but in certain aspects the description is *too narrow or too broad*, indicating that the selection of areas covered has not been undertaken consistently in relation to the purpose and other content. There are only *individual inadequacies* in how the results are dealt with, but the analysis and report give a balanced impression despite these inadequacies. b) The conclusions are clearly linked to the analysis of results and the purpose/problem formulation but *in some instances they have a different focus* from what is stated in the purpose/problem formulation. c) The question has been adequately processed. There is a clear connection between the research question, results and conclusions. The project's conclusions are substantiated and correct. d) Relevant knowledge and methodology have been *acquired and applied.* e) The project tackles the chosen area using *relevant, correct* language. The overall coherence, structure and layout are of *good* quality. f) The report includes a summary that reflects the entire content of the report in a *clear manner*.
BK	a) The subject description is *inadequate* because important areas have not been described. The results and analysis of results are not easily accessible on a first read. The selection is manifestly *poorly linked* to the purpose and general content, and in addition, there is *inadequate* balance in the report. b) *No (or very weak) link* and consistency between conclusions and purpose/problem formulation. Another problem may be that the purpose given in the introduction is not consistent with what is argued in the discussion/conclusions. c) The project *lacks or has an unclear* research question or set of objectives. Irrelevant methodology has been employed. The project does not provide an answer to the question or a result that relates to the objective. The conclusions are incorrect. d) Areas that are relevant to the project *are not considered or deployed*. Knowledge that has been chosen and acquired is not presented clearly and lacks justification. e) The project *largely lacks* adequate use of language, coherence, structure or layout to provide a context for the reader.

	f) The report *does not* include a summary that reflects the entire content of the report in a clear manner.

Specialisation

<table>
<tr><td colspan="2">The aim of a degree project is for students to develop specialised knowledge, understanding, skills and attitudes within the context of the programme. A degree project must be at the end of the programme and involve specialisation and synthesis of previously acquired knowledge.

The project must be based on relevant in-depth studies for the programme, and this should be clear from the report's theory and discussion as well as in the introduction where the students explain the purpose of the study. It shows that the students have read and absorbed literature and research that are relevant to the subject area.

NB! The criteria for specialisation in the project have been assessed to be such an important part that they earn double MHK or BK points towards the grade.</td></tr>
<tr><td>MHK (x2)</td><td>a) The theory chapter shows specialised knowledge of the subject. The sources are representative of the subject area. The problematisation of the research question is deep and demonstrates substantial knowledge of the subject.
b) Significant specialisation in the main field of study has been demonstrated. The project utilises knowledge of the main field of study.
c) A comprehensive review of existing literature and deep reflection that justifies the connection between the project and the forefront of knowledge in the main field of study.
d) The contribution to research or development work is clearly presented.</td></tr>
<tr><td>HK</td><td>a) The theory chapter demonstrates knowledge of the subject. The sources are representative of the subject area. The problematisation of the research question indicates good subject knowledge.
b) Specialisation in the main field of study has been demonstrated. The project utilises knowledge of the main field of study.
c) There is a written review of existing literature and reflection on the connection between the project and the forefront of knowledge in the main field of study.
d) The project presents a contribution to research or development work.</td></tr>
<tr><td>BK (x2)</td><td>a) The theory chapter is thin. The sources used are not representative of the subject area. The problematisation of the research question is superficial.
b) Specialisation in the main field of study is lacking or is vague.
c) The project's link to the main field of study is weak or lacking. The project does not utilise knowledge of the main field of study. A literature review and reflection on the project's link to the relevant knowledge area are lacking.
d) The project is such that it is difficult to link it to any research or development work.</td></tr>
</table>

Formalities and structure

<table>
<tr><td colspan="2">The report must follow the current template for degree projects.

To increase readability, the report must be well structured, and the style must be consistent. Chapters must be well balanced in length and content and have a clear, informative heading. References must be correctly managed and follow the chosen reference system throughout. Tables and figures must be in appropriate places in the report and be clear and referenced in the body text. Images, graphs, tables, etc. must comply with the Swedish Copyright Act. Sources must be stated clearly and correctly.</td></tr>
<tr><td>MHK</td><td>a) The report’s length and formalities comply with the writing instructions and are effectively adapted to the task. The formalities improve readability.
b) The whole report is divided into clear, well-balanced chapters, sections and paragraphs.
c) The referencing of tables, figures and sources is correct and does not affect readability. Any paraphrasing is very well balanced.
d) Tables and figures are in appropriate places in the text, have clear, informative table and figure headings and are well commented in the text.</td></tr>
<tr><td>HK</td><td>a) Length and formalities generally comply with the instructions but are not always effectively adapted and therefore affect readability. For example, the odd heading may not be particularly informative.
b) The report is generally divided into clear chapters, sections and paragraphs, but sometimes consists of very short or long chapters that affect the balance between the different parts.
c) The referencing is good but there are occasional inadequacies in the formalities. Several instances of clear paraphrasing but otherwise good referencing.
d) Tables and headings are in appropriate places in the text, but the table and figure headings are somewhat unclear. There are occasional tables and figures that are inadequately discussed in the text.</td></tr>
<tr><td>BK</td><td>a) Length and formalities do not follow the instructions and there are clear inadequacies, for example in the table of contents, layout and/or pagination.
b) Significant inadequacies in dealing with paragraphs, chapters and sections in terms of the scope of and relationship between the various parts.
c) Significant inadequacies in the referencing. For example, references to tables/figures are missing in the text, and/or several sources are missing in the bibliography.
d) Tables and figures are incorrectly placed and/or lack table and figure headings. Furthermore, explanatory comments are very often missing in the text.</td></tr>
</table>

Readability and language

<table>
<tr><td colspan="2">The criteria relating to language and style do not mention the concept of flow. This elusive aspect of text production is a result of factors such as language and style working together effectively with structure and content to create readability that is as effective as possible. Consequently, when inadequacies in flow are experienced, the reader sees more clearly which factors or aspects are lacking for the text to flow effectively.</td></tr>
<tr><td>MHK</td><td>a) Paragraphing is very good and follows the thought where there is one thought/idea per paragraph and the paragraphs have clear core sentences.
b) The sentence structure is correct, formal and effective (there are no incomplete sentences). There are only a few sentences with an information structure that is too complex (sentences that are difficult to read).
c) The report features correct choices of words while concepts fit together well and are used consistently.
d) The style level is even and does not fluctuate between formal and informal style. The report features good awareness of the applicable style standard.
e) The report has been carefully proofread and contains very few or no grammatical, construction or spelling errors.</td></tr>
<tr><td>HK</td><td>a) Paragraphing is good but there are certain inadequacies in that some paragraphs are short and poorly integrated with other paragraphs and/or some paragraphs contain several themes without having a core sentence to coordinate them.
b) The sentence structure is good and there are only a few examples of incorrect or inadequate sentence structure. Occasional sections have sentences with an information structure that is problematic.
c) Choice of words and concepts is dealt with adequately and appropriately.
d) The style of the report is at the correct level and the authors have managed the applicable style standard with few or no exceptions.
e) The report has been proofread and contains few errors at word and sentence level.</td></tr>
<tr><td>BK</td><td>a) There are significant inadequacies in the paragraphing in terms of both the formal introduction and the content of the paragraphs. The unclear paragraphing means that in several instances it is difficult to understand what a paragraph/section/chapter is about.
b) The text contains many instances of incorrect sentence structure, for example sentences without a subject and/or sentences that only consist of subordinate clauses. Several sentences contain personal pronouns (e.g. I, we, the undergraduates, etc.).
c) The report is inconsistent in terms of choice of words and concepts.</td></tr>
</table>

	d) There are *clear inadequacies* in the style of the report. The style level is uneven and several prominent breaches of style feature in the text. e) The report has *not been proofread* and contains so many spelling and typing errors throughout that this disturbs the readability of the text and makes it more difficult to read.

Introduction, background, purpose and theory

In the introduction, the student should cover the basic thinking behind the project and give reasons for why it is important without going into detail. The background should present the scientific work that has previously been performed and any other information that the reader needs for them to be able to understand the rest of the report. In addition, the project that is presented should be placed in a context to make clear that there is a unique contribution in the report. The purpose must show where this particular study is intended to lead. The purpose of the study is never that the students should pass the course or advance their own knowledge. The theory chapter must contain the theoretical basis the reader needs to understand the specific subject, either integrated in the background or as a separate theory chapter, depending on the scope and nature of the degree project. The theory chapter must be adapted to the target group.	
MHK	a) A *clear common thread* that enables the reader to easily and clearly understand the basic thinking behind the study, the reason for it and its unique contribution. b) The purpose of the study is *clear and well defined.* c) The theory is relevant in relation to the purpose and background, enables the reader to become familiar with the subject and is up to date with the support of relevant scientific literature. d) Scientific sources have been used, and for the most part consist of scientific articles and/or reports.
HK	a) The basic thinking, reason for the study and unique contribution *are not really coherent*. The reason for the study is unclear or the unique contribution is somewhat unclear. The common thread is inadequate or unclear in certain respects. b) The purpose of the study has been established but *the definition is somewhat wide/unclear*. c) The theory is relevant in relation to the purpose and background but is *too deep/superficial*. d) Scientific sources have been used.
BK	a) There is no common thread, and the background, purpose and theory *are not coherent*.

	b) The purpose is *unclear or incorrect* (the purpose is ‘to pass the course’ and similar). c) The theory is *not adequately* substantiated or is out of date. d) The references are not relevant or are missing.

Research question, method and results

<table>
<tr><td colspan="2">
The research question consists of a number of concrete questions to which the report should provide answers and is usually formulated as a main question and a number of sub-questions.
The chapter on methodology briefly explains how the choice of methodology is appropriate and reasonable in relation to the purpose and the research question without discussing it in detail. How the project was implemented is described in a clear, structured manner, who took part, how selections were made, time, place, etc.
The results chapter presents relevant data that was collected with the chosen methodology in a clear, structured manner without discussion in the case of qualitative methodology. The integration of results and analysis is permissible in quantitative studies. Interviews provide raw data which must be analysed in some way and not merely presented.
Within the framework of the specific degree project, the students must be able to identify research questions relating to the role of technology in society, taking into account environmental and ethical aspects.
</td></tr>
<tr><td>MHK</td><td>
a) The project has a clear, well-defined research question that comprises its entire purpose.
b) Potentially relevant engineering or scientific theories and methodology have been used. The choice of methodology is appropriate and reasonable in relation to the purpose and the research question.
c) The chosen methodology has been applied in a correct and/or innovative manner. The project demonstrates deep, broad methodological knowledge.
d) How the project was implemented is presented correctly, in a structured manner and in detail.
e) Very well conducted fact finding of high quality.
f) Results are presented correctly and well. With the selection of tables, figures and examples, the presentation of results as a whole is convincing.
g) The research question clarifies, if possible, the problem at hand in terms of the role of technology in society, taking into account environmental and ethical aspects.
h) The report is steeped in an ethically correct approach to oral sources.
</td></tr>
<tr><td>HK</td><td>
a) The project has a clear, well-defined research question, but this does not comprise the entire purpose.
b) Potentially relevant engineering or scientific theories and methodology have been used. The choice of methodology is appropriate and reasonable in relation to the purpose and the research question but there is better alternative methodology.
c) The chosen methodology has been applied in a correct manner. The project demonstrates a certain degree of methodological knowledge.
d) How the project was implemented is presented but there are certain inadequacies in terms of correctness, structure or detail.
e) Good fact finding but of somewhat variable quality and scope (though only individual and limited inadequacies).
</td></tr>
</table>

	f) The results are presented *largely correctly* and well, but individual inadequacies affect the project as a whole and its credibility. In addition, the interaction between text, figures and tables is *sometimes incomplete*. g) The research question *clarifies in part*, if possible, the problem at hand in terms of the role of technology in society, taking into account environmental and ethical aspects. h) The report is steeped in an ethically correct approach to oral sources.
BK	a) The research question in the project is *unclear and/or not well defined.* b) The theories and methodology selected for the project *lack relevance.* Inappropriate methodology in relation to the purpose/research question. c) The student has *not demonstrated* that they have a grasp of their chosen theories and methodology. d) How the project was implemented is *inadequate* and/or not sufficiently presented. e) Only a *small number of sources* have been used. Alternatively, the fact finding is of *inadequate quality*, for example because it consists of online sources with low credibility. f) *Incomplete and incorrect* presentation of results with significant inadequacies that are not limited to inadequate interaction between text, figures and tables. g) The research question *does not clarify* the problem at hand in terms of the role of technology in society, taking into account environmental and ethical aspects, although it would be possible to do so. h) The report is not steeped in an ethically correct approach to oral sources.

Discussions and conclusions

The results presented in the results chapter must be discussed in relation to theory and previous research in the discussion chapter. This chapter provides space for the author's own thinking, but this must not be too speculative in relation to theory, previous research and results. It is important for data and results to be critically reviewed and discussed. The student must also, via a discussion of methodology, discuss the advantages and disadvantages of the methodology as well as the methodology's credibility (the concepts of validity and reliability in appropriate instances). Based on the results that are produced and the discussion that has taken place, conclusions must be formulated in a relatively short chapter in which an answer is given to the research question as formulated, if this can be demonstrated, and provide suggestions for further work to be done in the area. The discussion chapter must also, where possible, discuss the results on the basis of the role of technology in society, taking into account environmental and ethical aspects.	
MHK	a) The argumentation is *objective, well structured and well balanced*. It is substantiated by correct content and relevant references, and is supported by clear examples and by the results presented.

	b) The results are analysed and evaluated critically. *Alternative hypotheses have been developed* and treated in a relevant manner. c) The methodology discussion contains the advantages and disadvantages of the chosen methodology in relation to results and *any alternative methodology.* The credibility of the methodology is critically discussed. d) The project's conclusions are *well substantiated* and clear, and provide answers to the study's research question. e) The role of technology in society, taking into account environmental and ethical aspects, *is discussed (in appropriate instances)*.
HK	a) The argumentation is *objective but with certain inadequacies* in terms of structure and support for the arguments that are put forward. The authors are able to adapt the strength of various statements to take the line of argumentation forward. b) The results are analysed and evaluated critically. c) The methodology discussion includes the chosen methodology's advantages and disadvantages in relation to the results. The credibility of the methodology is discussed. d) The conclusions are *substantiated* and largely provide answers to the study's research question. e) The role of technology in society, taking into account environmental and ethical aspects, *is discussed superficially (in appropriate instances)*.
BK	a) The argumentation is *poorly substantiated.* The statements lack support and include too many categoric statements. b) The results are not analysed and evaluated critically. c) The methodology discussion does *not* include the chosen methodology's advantages and disadvantages in relation to the results. The credibility of the methodology is not discussed. d) The conclusions are *inadequately substantiated* and/or do not provide answers to the study's research question. e) The role of technology in society, taking into account environmental and ethical aspects, *is not discussed (in appropriate instances)*.

Oral presentation and opposition

Oral presentation and opposition must be in accordance with instructions. The oral presentation must be sufficiently detailed for students on the same programme who have not read the report to be able to follow the presentation and also take part in the subsequent discussion. Visual aids must be used to illustrate the presentation. It is important for the presentation to have relevant content, be adequately comprehensive and be presented such that the message reaches the audience.

Well conducted opposition features relevant questions, the following up of questions and the ability to create context for the audience. The opposition must be introduced and concluded in a well thought out way and the content must be well adapted to the communication situation. Personal attacks and abuse will result in an immediate fail grade. Presentation: • A *good selection* of material from the report and the degree project in its entirety • The project is presented in a form that is *well adapted* to the recipients, the situation and the specific subject area • *The content is well structured,* and the presentation is therefore easy to follow. The introduction and conclusion are clearly marked and help the recipients absorb the content. The different parts are well linked and together they form a unified whole • The *visualisation material* (images, graphs, bullet lists, text) used is clear and does not contain too much information. The material is readily comprehensible • The presentation is kept within the given *time frame* Oral opposition: • The *oral opposition* features relevant questions, the following up of questions and the ability to create context for the audience. • The *oral opposition* promotes discussion and is not a detailed scrutiny Group criteria (not applicable to individual projects): • *The handover* between different speakers and between sections is well planned and does not result in any disruptive interruptions in the presentation • *The division* between different group members is relatively equal in terms of both speaking time and answering questions Individual criteria: • Each individual speaker in the group establishes and maintains *good eye contact* with the audience and *speaks extemporaneously*, with memory aids • *The speaker* presents the material shown in clear, comprehensible terms. The presentation of the visual material is logical and well conceived • *The performance is engaging and arouses interest* • The individual *responds* well to relevant questions • The *written opposition* is a balanced and well-formulated critique of the report in question and how well the opponents were able to familiarise themselves with it

MHK	12–14 criteria met (individual project: 10–12)
HK	5–11 criteria met (individual project: 4–9)
BK (= new presentation required)	0–4 criteria met (individual project: 0–3)

Grading criteria

Grade	Criteria
5	At least 7 MHK, no BK
4	At least 4 MHK, no BK
3	No BK
K (Supplementary work to grade 3 may be done)	1–3 BK
U (The project must be redone on a new subject)	≥ 4 BK

APPENDIX 4. Guiding principles for assessing the quality of degree projects on Chalmers' Business Development and Entrepreneurship programme

The guiding principles are specifically intended to be a guide in cases in which the quality of the degree project is low, and there is reason to consider whether it can be passed. The guiding principles may also be a support when giving feedback to the student on their performance.

The guiding principles are based on the learning objectives for degree projects in Bachelor's programmes.

Criteria for poor quality/high quality are formulated for all learning objectives. Criteria for Very High Quality (MHK) are formulated only for specific learning objectives, those that are judged to be of a distinctive nature: specialisation in the technical area, problem formulation, ability to create and evaluate new solutions, integration of knowledge, written presentation and independence.

Poor quality in terms of a learning objective or several learning objectives in combination may result in a fail grade for the entire project. It is the examiner's responsibility to make a balanced assessment.

Criteria for Poor Quality (BK), High Quality (HK) and Very High Quality (MHK) for each of the learning objectives for the degree project are shown below.

1. **Ability to acquire and apply specialised knowledge in the technical area of the study programme, including specialised insight into current development work:**

MHK	Specialised knowledge in the main field of study is demonstrated. The project contributes, in a clearly presented way, to new knowledge or new application in the

	main field of study. The project demonstrates the ability to work independently in the field.
HK	Specialised knowledge in the main field of study is demonstrated. The project shows the ability to work independently in the field.
BK	The project does not utilise specialised knowledge in the main field of study. The project's link to the main field of study is weak or lacking.

2. **The ability to independently and creatively identify, formulate, and manage questions with a holistic perspective, and analyse and evaluate various technical solutions:**

MHK	The project has a clear, distinct question or set of objectives. The question/objectives have been processed in an adequate, critical and reflective way. New solutions are developed and analysed and evaluated critically in the project. Alternative solutions have been developed and processed in a relevant, exhaustive manner. There is a clear connection between question/objectives, results, discussions and conclusions. The project's conclusions are well substantiated and correct.
HK	The project has a clear, distinct question. The question has been adequately processed. Solutions are developed, analysed and evaluated critically in the project. The project's conclusions are well substantiated and correct. There is a clear connection between question, objectives, results and conclusions.
BK	The project lacks or has an unclear question or set of objectives. Irrelevant methodology has been employed. Alternative solutions have not been identified. The project does not provide an answer to the question or a result that relates to the objective. The conclusions are incorrect.

3. **Ability to plan and carry out tasks with adequate methods within given frameworks:**

HK	A realistic plan has been drawn up for the project. The deadlines that were communicated and established have been adhered to during the implementation of the project. Any adjustments that have been necessary for the implementation have been documented and communicated.
BK	The project has not adhered to the deadlines communicated and established. Furthermore, documentation of relevant factors for deviations has not been provided.

4. **Ability to critically and systematically apply knowledge, and to model, simulate, predict and evaluate phenomena based on relevant information:**

MHK	The project integrates knowledge and methods from several subjects.
HK	Relevant knowledge and methods have been acquired and applied.
BK	Areas of relevance to the project are not addressed or used. Knowledge that has been chosen and acquired is not presented clearly and lacks justification.

5. Ability to give an account of and discuss information, problems and solutions orally and in writing:

MHK	A well-written report. The overall coherence, structure and layout are of high quality.
HK	The project addresses the chosen area with relevant, accurate use of language. The overall coherence, structure and layout are of good quality.
BK	The project primarily lacks adequate use of language, making it difficult to understand or assess the project based on the report.

The guiding principles for the assessment of degree project reports can provide guidance for the assessment of the written report. See the document Assessment of the written presentation, report - HISS (version 2011-01-10).

http://www.ait.gu.se/facksprak/kandidatarbete/skrivanvisningar/

6. **Ability to identify, within the scope of the specific degree project, questions regarding the role of technology in society, such as environmental and ethical aspects:**

HK	Presents and justifies chosen methods and discusses results from a perspective with a focus on sustainable development. Presents possible ethical consequences of the project performed.
BK	Does not consider these aspects even though the examiner deems them to be of importance for the degree project in question.

This learning objective may be irrelevant for certain degree projects. This assessment is made by the examiner.

7. **After having completed the degree project, the student must have demonstrated the knowledge and skills required to work independently after having graduated with a Degree of Bachelor of Science:**

MHK	A degree project implemented independently and without extraordinary support, or any other significant extra resources for its implementation.
HK	Implemented the project with reasonable support.
BK	There has been a great need for support. This support has been too extensive to make it likely that the student will be able to work independently after graduation.

Degree project risk assessment	CHALMERS	
Educational programme:	Date:	Department:
Name of student:	Name of examiner:	Name of external supervisor:
Risk-assessed degree project (title):		The degree project is being implemented at (company name/workplace/Chalmers):

The risk assessment has been performed in consultation with an external supervisor: ☐

Explanation
This template is used to risk-assess the project and the steps/labs/experiments that will be carried out during the degree project. The purpose of the risk assessment is to identify and prevent potential risks to create a safe working environment for all involved. By identifying potential hazards in advance and taking appropriate safety measures, accidents and near-misses can be avoided.

The risk assessment must be performed with the examiner and any external supervisor. The risk assessment must be attached to the planning report and approved by the examiner.

How to fill in the template:
- In the ‘Source of risk’ column, describe the risk identified in as much detail as possible
- Then assess the severity of the risk.
- In the ‘Action’ column, describe how the risk will be eliminated or reduced.
- In the ‘Responsible’ column, enter the name of the person responsible for taking the risk reduction measure.
- In the ‘Done by’ column, specify when the actions must have been implemented.
- In the ‘Result’ column, document the actions that have actually been implemented so that it is clear what has been implemented in practice and not just planned.

Remember that if steps being risk-assessed will be performed several times, it is important to look at the risk assessment in between the steps and update it if you think something did not work.

Source of risk	Risk assessment			Action	Responsible	Done by?	Result
	low	med	high				
	☐	☐	☐				
	☐	☐	☐				
	☐	☐	☐				
	☐	☐	☐				
	☐	☐	☐				

Source of risk	Risk assessment			Action	Responsible	Done by?	Result
	low	med	high				
	☐	☐	☐				
	☐	☐	☐				
	☐	☐	☐				
	☐	☐	☐				